

“PET Project” Proposal

Touch Designer Friends: Elisa Lupin-Jimenez, Tala Salman, Paola Rodriguez Vitale

Introduction

Objective

Map embodied motion of a user to dynamic visuals rendered in TouchDesigner.

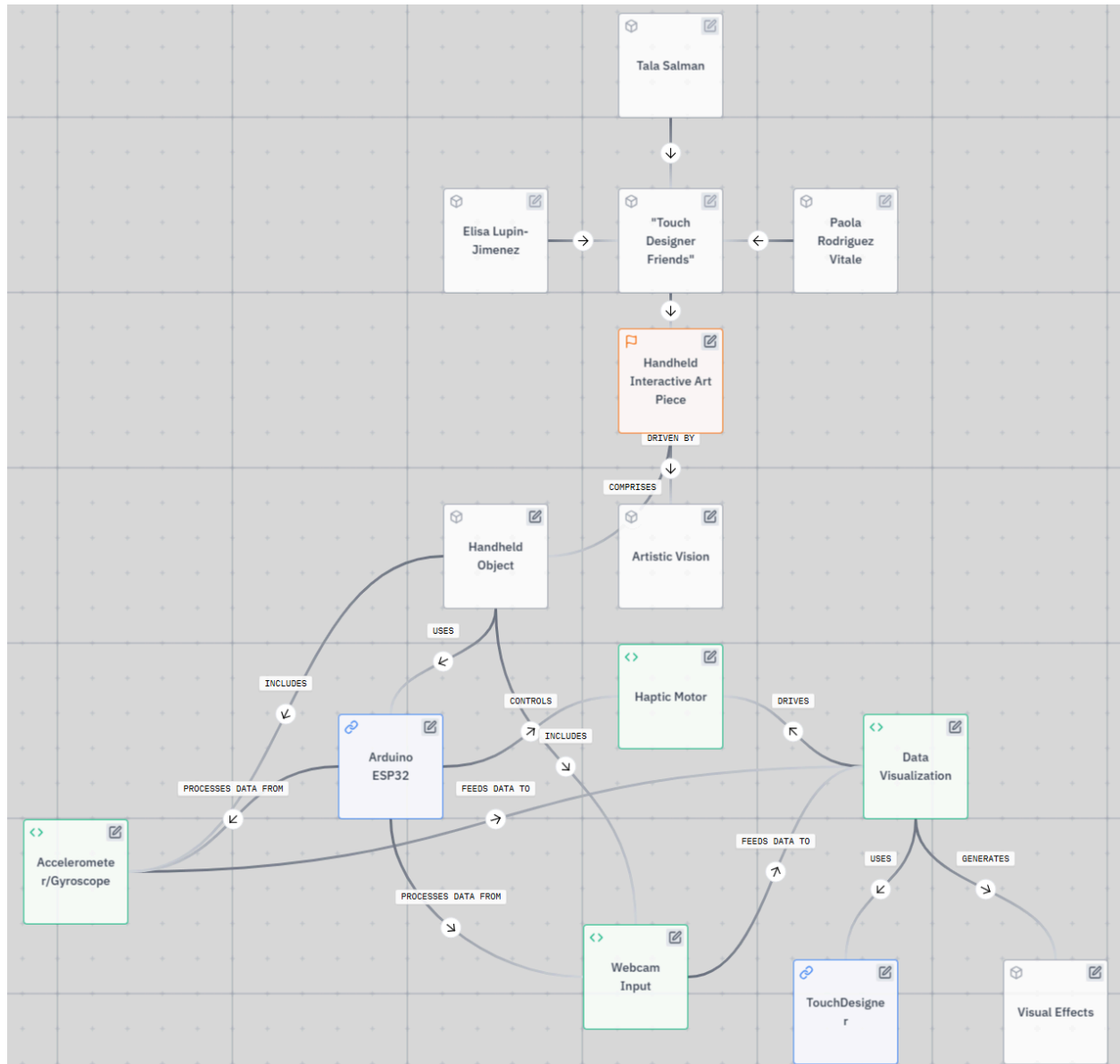
Components

- Screen
 - Monitor
 - Projector (optional)
- TouchDesigner
- Computer vision
 - p5.js
- Arduino
 - ESP32
- Electronic components
 - Gyroscope
 - Accelerometer
 - Haptic motor

Scenario

- A single user regulates their stress through intentional movement with our device that provides engaging visual feedback.
 - An artist uses our device as a medium to convey synesthetic movement to an audience.
 - An installation device used for meditation.
 - A collective or individual biofeedback installation.
 - Multiple users collaboratively play with our device (or multiple devices) to “paint” on a digital canvas.
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System Architecture



Experimentation Plan

Experiment 1: IMU calibration + filtering

Components

- Arduino ESP32
- IMU component
 - Gyroscope
 - Accelerometer

Objective

- Clean, stable orientation / motion features from the IMU on ESP32

Focus Area

- Fine-tune sensitivity of movement sensors

Dependencies

- Supplying power to the device

Steps

1. Wire IMU → ESP32; run sample read.
2. Capture raw accel/gyro at rest and during slow rotations ($\pm 90^\circ$).
3. Compute bias/offset (mean at rest), remove gravity from accel.
4. Add filtering: start with moving average; if time allows, complementary filter (gyro drift + accel gravity).
5. Export features: roll, pitch, |accel|, |gyro|, jerk (Δ accel).

Experiment 2: Testing the serial communication between TouchDesigner & ESP32

Components

- ESP32
- Laptop with TouchDesigner installed

Objective

- Loss-free, low-latency stream from ESP32 → TouchDesigner.

Focus Area

- Create a reliable line of communication from ESP32 to TouchDesigner

Dependencies

- ESP32 connected to WiFi

Steps

1. Define a compact packet schema (e.g., #t, ax, ay, az, gx, gy, gz\n) + checksum or line-based validation.
2. Test baud at 115200 and 9800; measure packet loss in Python or TD.
3. Implement simple re-send on invalid checksum (optional).
4. In TD, use Serial DAT → DAT to CHOP → Rename/Math CHOP.

Experiment 3: Mock ESP32 sensor data to TD and design a dynamic visual

Components

- Computer with TouchDesigner
- Mock output from an IMU (string)

Objective

- Design and test one responsive visual state in TouchDesigner using simulated accelerometer/gyroscope data, before connecting the physical ESP32.

Focus Area

- Build a minimal TD patch that accepts fake motion data.
- Map one variable (e.g., overall |accel| or tilt angle) to a visual behavior (brightness, hue, or noise intensity.)
- Tune filtering and scaling for smooth, continuous response.
- Evaluate the visual's emotional tone and responsiveness.

Dependencies

- Correctly formatted mock data
- Working Serial or OSC connection to TD
- Functional TD node chain.

Steps

1. Generate mock sensor data stream using code or pre-recorded file.
2. Import data into TD
3. Map one data parameter to control a single visual property.
4. Adjust scaling/filtering for fluid transitions.

Experiment 4: Building the visual interface and Haptic Feedback

Components

- TouchDesigner
- Arduino IDE
- Haptic motor

Objective

- Develop the visual identity and interaction logic for the installation in TouchDesigner.
- Create and test one steady haptic pattern

Focus Area

- Build a visual composition that can later react to movement data (from the IMU).
- Design one consistent pulse pattern (e.g., slow, even vibration).
- Control intensity and rhythm using PWM.
- Check comfort, safety, and stability.

Dependencies

- Stable TouchDesigner patch structure for later sensor integration.
- Sufficient GPU performance for real-time rendering.

Steps

1. Prototype multiple visual sketches in TD (using Noise, Feedback, and Level TOPs.)
2. Tune parameters such as brightness, hue shift, and feedback delay to craft one cohesive visual atmosphere.
3. Add interactive controls (sliders, buttons) to simulate real-time modulation.

Experiment 5: Alternative Motion Input Testing

Components

- Laptop or camera-enabled device
- Motion tracking software (p5.js)
- TouchDesigner

Objective

- Explore a hands-free input method for motion detection that can serve as a backup or complementary data stream to the IMU.

Focus Area

- Establish consistent real-time communication between an external motion input and TouchDesigner's visualization system.

Dependencies

- Stable network connection
- Adequate lighting and camera visibility

Steps

1. p5.js sketch: calculate frame-diff motion energy (sum of absolute diffs).
2. Send value via OSC/UDP (node bridge) to TD.
3. In TD, treat it like another channel (A/B input selector).

Experiment 6: Physical Form - Enclosure Prototype Test

Components

- ESP32
- IMU sensor (accelerometer/gyroscope)
- Laser cut or 3D-printed shell
- Soft foam insert or cardboard material
- Wires, USB cable, small battery

Objective

- Design and test a stable, ergonomic form that securely houses the ESP32 and IMU, reducing motion noise while maintaining comfortable handling.
- Build and test a prototype enclosure that conceals electronics, organizes wiring internally, and provides a smooth, ergonomic experience.

Focus Area

- Evaluate the physical design's effect on sensor noise and usability; compare vibration damping and comfort across prototypes.

Dependencies

- Working ESP32 + IMU circuit
- 3D printer or physical modeling materials

Steps

1. Quick 3D print or laser cut form to explore grip + weight.
2. 3D print simple shell (two halves), add soft insert for IMU isolation.

3. Internal wire-management; strain relief; charging access.
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Team Member Roles

Each member will take the lead in the skillset they came into the project with. Their responsibility includes bringing the team up to speed on that skillset.

Elisa Lupin-Jimenez

Lead

- Input Layer (Sensors & Data Capture)

Responsibilities

- Gyroscope/accelerometer input to ESP32
- Investigate TouchDesigner integrations
- Software development on ESP32 and TD communication
- Work with Tala and Paola to learn CAD (AutoCAD, Rhino, etc.) to build the physical device
- Work with team on software development

Motivation

- Learn how to use TouchDesigner and create a connection between hardware and software to create hybrid immersive technologies.

Tala Salman

Lead

- Processing Layer (Interaction & Feedback)

Responsibilities

- Haptic output component with ESP32
- Investigate TouchDesigner Visuals
- Interaction design
- Interface storyboarding
- Work with Paola to embed the haptic motor in the physical form safely
- Work with team on CAD prototyping

Motivation

- Learn TouchDesigner while researching how interactive systems influence emotional regulation and connection.

Paola Rodriguez

Lead

- Form & Redundancy Layer (Physical Design + Backup Input)

Responsibilities

- Develop the p5.js motion-tracking system communicating with TouchDesigner (via OSC).
- Design and prototype the physical form integrating sensors and haptics.
- Lead testing on data reliability, sensor stability, and user ergonomics.
- Support concept development and documentation of system behavior.

Motivation

- Learn new tools like TouchDesigner, projection mapping and explore how biofeedback can create new forms of interactions between people and technology.
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Expected Outcomes & Initial Feasibility Metrics

Expected Outcomes

1. **Working Prototype:** A handheld object that connects motion data to responsive visuals in TouchDesigner.
2. **One Haptic–Visual State:** A calm, grounded feedback mode where vibration and light work together to guide focus.
3. **Embodied Interaction Framework:** Early mappings between movement, emotion, and sensory feedback that can evolve into future states.
4. **Reflections on Emotion + Motion:** Insights gathered from testing how people's gestures and rhythms shift when feedback is calm, not commanding.
5. **Showcase-Ready Demo:** A small, self-contained installation that feels personal, poetic, and technically coherent, ready to share and iterate on.

Feasibility Metrics

1. IMU calibration + filtering
 - a. Success Metrics
 - Sensor accurately detects and reports changes in orientation and speed
2. Serial Communication (ESP32 → TouchDesigner)

- a. Success Metrics
 - End-to-end delay (movement → screen change)
 - TouchDesigner keeps a stable framerate while receiving live data.
- 3. TouchDesigner Visual Mapping
 - a. Success Metrics
 - Mock (fake) sensor data creates a smooth, natural visual response (no sudden jumps).
 - The visual clearly shows a specific emotion or energy (like calm or active).
- 4. Haptic Feedback
 - a. Success Metrics
 - The intensity of the TouchDesigner visual maps to the intensity of the haptic feedback.
- 5. Webcam-motion fallback via p5.js
 - a. Success Metrics
 - Webcam detects changes in body movement and positioning reliably
 - E.g., hands raised above shoulders vs. below shoulders
- 6. Handheld Form + Sensor Mounting
 - a. Success Metrics
 - All electronic components are fully enclosed and hidden from view.
 - The form feels cohesive, comfortable, and durable when handled.
 - No visible wiring or loose elements disrupt the interaction experience.

Timeline

November 3rd 2025 - December 9th 2025

Week 1: 11/3 - 11/8	Research: • Initial research • Proposal drafting
Week 2: 11/9 - 11/15	Ideation: • Submit proposal • Field exploration of data visualization • TouchDesigner workshops and training • Feasibility testing ◦ IMU calibration + filtering ◦ Set up serial communication between TD and ESP32 ◦ Real-time TD mapping (sensor → visuals)

	○ Haptic feedback (vibe motor mapped to motion state) ○ Webcam-motion fallback via p5.js ○ Handheld form mock + sensor mounting
Week 3: 11/16 - 11/22	Concept Development & Prototyping: ● Full integration of hardware components and software ● Prototyping of physical form (3D printing, laser cutting) ● Refinement of TouchDesigner / hardware mapping interaction
Week 4: 11/23 - 11/29	Testing: ● Final prototyping of physical form (3D printing, laser cutting) ● Refining TouchDesigner projection mapping to physical movement
Week 5: 11/30 - 12/6	● Preparation for in-class final critique ● Documentation clean-up
Week 6: 12/7 - 12/9	● Final revisions for Jacobs showcase

Summary

This project focuses on creating a single-user interactive system that uses motion sensors and TouchDesigner to generate real-time visual and haptic feedback. The goal is to explore how physical movement can influence a user's emotional state through visual projections. By linking gesture data to responsive visuals and gentle vibration, the system helps users observe the connection between their body and mind. The final outcome will be a working prototype that demonstrates how interactive media can support personal focus and calmness.

Inspiration

- [Visualizing mental health with TouchDesigner](#)
- [Generative laser production with TouchDesigner](#)
- Heartbeat Topography
 - <https://www.behance.net/gallery/172969237/Heartbeat-Topography>
- Contact juggling (for form)
 - <https://youtu.be/0rqeR-iLKYA?si=-iWfEhF5ostxW2gl>
- Multisensory movement space

- <https://derivative.ca/community-post/tabula-rasa-multisensory-expression-space-autism-foundation-finland-helsinki/69936>
- Microdose VR (matching movement to dynamic visuals)
 - <https://youtu.be/lKpidtxtst4?si=wyPpogAccFkgzbE4>
- Breathing visual simulator mobile app
 - <https://www.lungy.app/>